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PROJECT AND TEAM INFORMATION

Project Title

EV-OPT: Intelligent EV Charging Station Recommendation and Smart Charging Management

Team Name: Team #	EV-OPT T097
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

EV-OPT is an innovative electric vehicle charging management system. The aim of EV-OPT is to help electric vehicle users find the suitable charging station and manage charging efficiently. It takes into account factors such as distance, travel time, waiting time, charger availability and station conditions. This helps users choose a station that fits their needs. The project is built using C++. Follows object-oriented programming principles. It includes classes such as User, EV, Station, Charger, Charging Request, Reservation and Charging Session. Data structures like graphs, queues, priority queues and hashing are used to handle routing, manage requests and quickly find records. Currently the project is, in the integration stage, where all these parts are being connected to build a charging management system.

Updated Project Approach and Architecture (2 pts)

EV-OPT is divided into modules

First the user enters details like location, destination and battery level. The system then studies charging stations and ranks them based on distance, travel time, waiting time and charger availability. Dijkstra's algorithm is used to find the route. The project uses C++ and object-oriented programming with classes, like User, EV, Station, Charger, Charging Request, Reservation and Charging Session.

After station is selected, the charging requests enter a priority queue. The scheduler selects requests based on their priority. Finally, the available station power is allocated to charging sessions while following power limits.

Tasks Completed (7 pts)

Task Completed	Team Member
Basic C++ classes for User, EV, Station, Charger, Request, Reservation and Session	Member 1
Graph structure and Dijkstra approach started	Member 2
Recommendation factors finalized	Member 3
Charging priority and power allocation approach designed	Member 4

Challenges/Roadblocks (7 pts)

The main challenge was connecting all the project modules so they could work together properly. Another challenge was finding the best charging station by considering factors other than distance, such as waiting time and charger availability. We also needed a way to manage multiple EVs requesting charging at the same station. To solve this, we planned to use a priority queue and a power allocation system to manage charging requests and available power efficiently. We have selected a priority queue to handle urgent requests before others. The other problem is power allocation. A station has limited power. The system is built to distribute available power over active charging sessions without passing the station limit.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
Connect all modules into the main system	Member 1
improve route calculation with project inputs	Member 2
Connect recommendation with the user flow	Member 3
Connect priority queue with charging requests.	Member 4
Complete final testing and debugging.	Member All Members

Project Outcome/Deliverables (2 pts)

EV-OPT is a software prototype that helps EV users find suitable charging stations and manage the charging process efficiently. In Phase II, the main application is being developed in C++ using Object Oriented concepts. The system consists of several classes, including User, EV, Station, Charger, Charging Request, Reservation, and Charging Session. The recommendation module considers factors such as travel time, waiting time, charger availability, station occupancy, charging cost, and the urgency of EVs to suggest suitable charging stations. The reservation and charging session modules handle bookings and manage charging activities. Meanwhile, the smart charging module prioritizes charging requests to ensure efficient use of the available chargers.

Progress Overview (2 pts)

During Phase-II, **The C++ OOP layer with classes for User, EV, Station, Charger, Charging Request, Reservation and Charging Session was developed**. The system is able to receive EV and user data, generate requests, assign them to stations, create reservations and initiate charging sessions. The recommendation and routing components were conceptualized as the station-selection layer whereas the smart charging was implemented as the request-level management **using both queue- and priority-based approaches**. The work during Phase-II therefore created the foundation for the application and its core charging-management functionality. **The UI layer**, automatic station status updates, **full session lifecycle**, analytics, testing and final system bring the project to its conclusion **during Phase-III**.

Codebase Information (2 pts)

Repository: EV-OPT

GitHub Link: [<https://github.com/itssarthakv/EV-OPT>]

Branch: 'main'

The codebase contains modular C++ files for user, EV, station, charger, reservation, and charging-session management. Scheduling and power allocation are under development.

Testing and Validation Status (2 pts)

Test Type	Status (Pass/Fail)	Notes
C++ Compilation	Pending	Build verification
User Validation	Implemented	Testing pending
Reservation Session	Implemented	Testing pending
Scheduler Power Allocation	Pending	Development
Frontend Integration	Pending	Development

Deliverables Progress (2 pts)

Completed: Core C++ classes, user input, Gmail format validation, station/charger management, reservations, and basic charging-session simulation.

In Progress: Priority scheduling and power allocation.

Pending: Station recommendation, web dashboards, persistent storage, analytics, and end-to-end testing.